The empirical recurrence for OEIS A282646: recovering a conjecture that is not written down

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2 September 2026

Abstract

OEIS A282646 records "Empirical recurrence of order 66", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 86$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 66, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 66 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A282646 is "Number of nX7 0..1 arrays with no 1 equal to more than one of its king-move neighbors.". It has offset 1 and begins

$$81, 1444, 25153, 557439, 10614316, 213392087, 4257307148, 84514081303, \ \ldots$$

The entry states:

> Empirical recurrence of order 66 (see link above)

As of the "Last modified" line on the live entry (Feb 20 2017 07:32:27, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1525$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 86$ of the 1525, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 86$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 172$ exact terms of the model returns order 66 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{66} c_i\, a(n-i),$$

c_1	8	c_{18}	−1319268700965	c_{35}	518131461823909	c_{52}	2651282262992
c_2	166	c_{19}	1412263129877	c_{36}	−257641062704913	c_{53}	40192623018
c_3	1720	c_{20}	−4820039058641	c_{37}	−607991043167863	c_{54}	284095497383
c_4	−3988	c_{21}	8200165212330	c_{38}	−205635415324467	c_{55}	43925632901
c_5	−26860	c_{22}	−4558447953296	c_{39}	129898736752336	c_{56}	11371807287
c_6	−335895	c_{23}	27584908491840	c_{40}	385594539300860	c_{57}	−10344337436
c_7	1453961	c_{24}	−27281656253651	c_{41}	41638837995277	c_{58}	−12286963896
c_8	−7620910	c_{25}	6422380788974	c_{42}	44979860011612	c_{59}	−1248830072
c_9	69071028	c_{26}	−106290381870888	c_{43}	−222268259619529	c_{60}	−160068002
c_{10}	−236382690	c_{27}	51591743953262	c_{44}	−42559629011401	c_{61}	426405936
c_{11}	978810310	c_{28}	16948766714914	c_{45}	−82376334016440	c_{62}	83460610
c_{12}	−4481669519	c_{29}	254662828297309	c_{46}	58118371143267	c_{63}	13412988
c_{13}	11442817948	c_{30}	−127491122140228	c_{47}	−32234396052725	c_{64}	−6709936
c_{14}	−37757480670	c_{31}	−230551336257235	c_{48}	4247124213492	c_{65}	−1758672
c_{15}	118035002789	c_{32}	−478334974221487	c_{49}	−20458132409853	c_{66}	80960
c_{16}	−203098725108	c_{33}	234933261728971	c_{50}	4784041033563		
c_{17}	569035914512	c_{34}	580728907503218	c_{51}	−3564710452485		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 66, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $66+4$ terms removed returns order 66 as well, so $\deg q_{\min} = 66 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $66+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 66 that this sequence satisfies — and that family turns out to have exactly one member.

References

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